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Paleomagnetic study of 1765 Ma dyke swarm from the Singhbhum Craton: Implications to the paleogeography of India

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ABSTRACT

We report the first key paleopole as a result of paleomagnetic study on a precisely dated 1765.3 ± 1.0 Ma WNW-ESE trending dyke swarm from Singhbhum Craton. This pole has been used in this study to propose the paleogeographic reconstruction of India with Baltica Craton and North China Craton. Incremental alternating field (AF) and thermal demagnetization, isolated high coercivity components with north to north-westerly declination and shallow negative inclination from 9 sampling sites which are representing different individual dykes. The primary origin of the ChRM is supported by the positive baked contact test. The WNW-ESE trending dykes yield a mean paleomagnetic direction with a declination = 329.2° and an inclination = −22.8° (k = 31.6; α95 = 9.3°). The positive bake contact test proves the primary nature of remanence. The pole position of Singhbhum Craton at 1765 Ma is 45°N, 311°E (dp = 5.2 and dm = 9.9). Paleogeographic reconstruction at ca. 1770 Ma, supported by geological, tectonic and metallogenic evidences indicate that the Baltica Craton and India linkage can be stable for at least ~370 Ma (~1770–1400 Ma). There is also reasonable evidence in support of India-North China Craton spatial proximity at ~1770 Ma.

1. Introduction

The paleogeographic position of India within pre-Rodinia supercontinent, variously called Nuna, or Columbia, or Hudsonland, has been a topic of debate during the last decade. There is a lack of consensus over the probable neighbours of India during Paleo-Mesoproterozoic. Among various configurations proposed for pre-Rodinia supercontinent, India was placed juxtaposed to south-west Baltica Craton (Pisarevsky et al., 2013, 2014); at equator separated from the amalgamation of Laurentia, Baltica and Siberia (Belica et al., 2014); next to northern boundary of North China craton (Zhang et al., 2012); and in close proximity to southern margin of the North China Craton (Xu et al., 2014). The position of western India juxtaposed to south-west Baltica (Pisarevsky et al., 2013, 2014) was largely based on the 1465 Ma Lakhna dyke paleopole and geological correlation between Dharwar Craton and Sarmatia Craton. This India-Baltica reconstruction can also be tested at ~1770 Ma with the help of 1765 Ma key paleopole of this study with coeval poles from Baltica Craton.

Western India was placed next to northern boundary of North China Craton (Zhang et al., 2012). India was placed in close proximity to southern margin of the North China Craton by Xu et al. (2014). The above mentioned India-North China Craton reconstructions have used 1798 Ma Gwalior trap pole. We would like to point that the Gwalior Traps were not precisely dated (1798 ± 120 Ma) and there is no field test available on Gwalior trap to prove the primary nature of remanence. This makes Gwalior trap pole unreliable and the ~1770 Ma reconstructions for India-North China Craton (Zhang et al., 2012; Xu et al., 2014) speculative. Here also 1765 Ma key paleopole of this study can be used to perform paleogeographic reconstruction with coeval poles from North China Craton.

In this study we are presenting the first key paleopole at 1765 Ma on WNW-ESE dyke swarm from Singhbhum Craton (SC). This pole will be used to explore the possible paleogeographic relation of India with Baltica Craton at 1770 Ma, to test the stability of proposed India-Baltica assemblage by Pisarevsky et al. (2013, 2014). This pole will also be used for paleogeographic reconstruction of India with North China Craton at ~1770 Ma.

2. Geology

The SC is about 40,000 sq. km. in extent in eastern India and lies between latitudes 21°00′ and 23°15′ and longitudes 84°40′ and 86°45′ (Saha, 1994; Fig. 1). The earliest work on SC was by Voysey (1844) and then onwards this area has attracted the attention of several workers.

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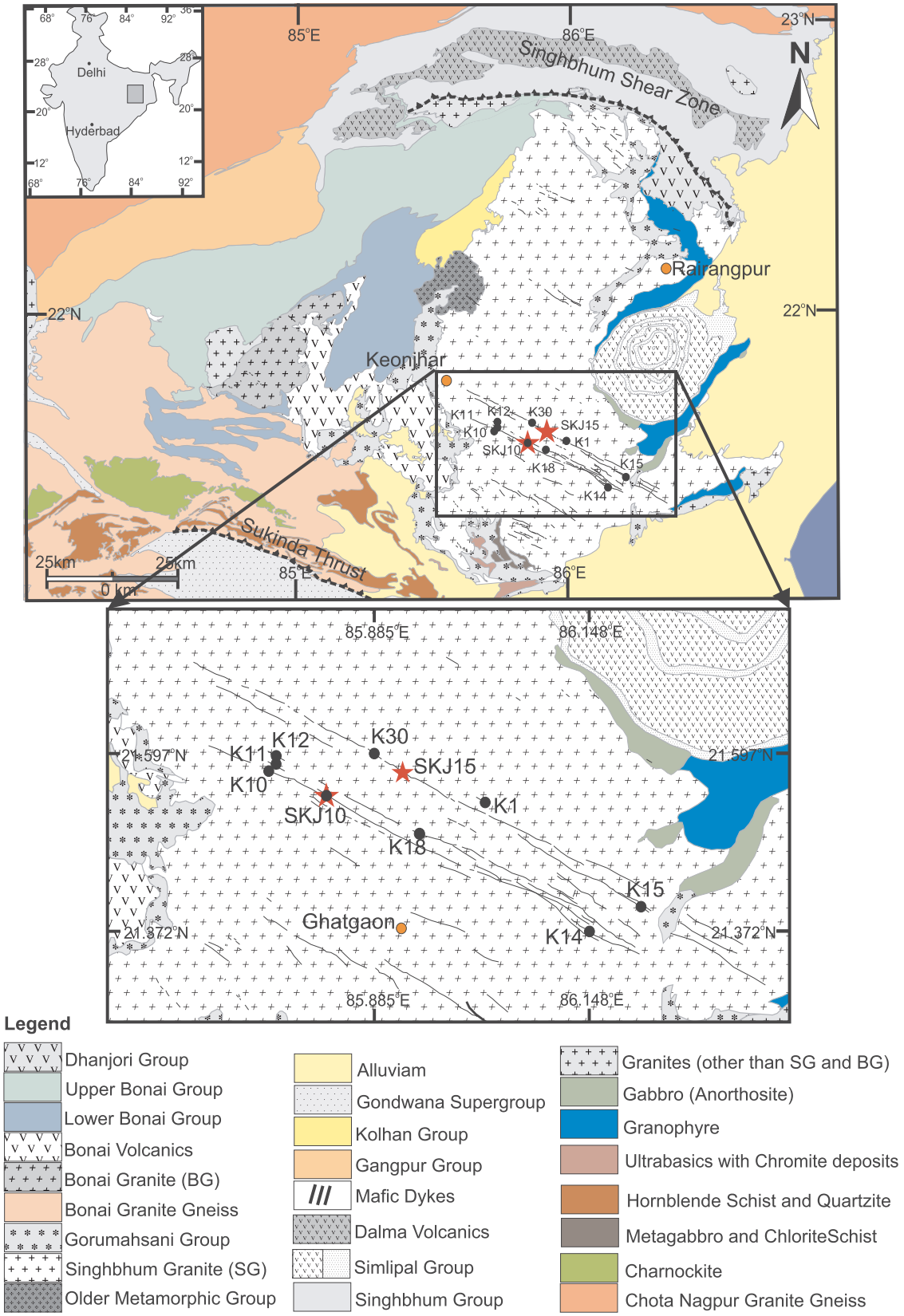


Fig. 1. Simplified geology map of a part of Singhbhum Craton (adapted from the Geological Survey of India map (GSI, Geological Map of India, 1:2,000,000 scale. Geological Survey of India, Hyderabad, 1998, and dyke distribution after Google Earth satellite images) showing the sample locations (enlarged view in inset shows the sampling locations on distinct dykes). Stars indicate the dated published dyke locations and black solid circle indicate the sites of paleomagnetic study from present study.

The area under study is dominated by granitic rocks (popularly called as Singhbhum granite). The Singhbhum granites were intruded extensively by the mafic dykes swarms (popularly called as Newer Dolerites Dyke Swarm (NDDS) (Dunn and Dey, 1942)). The NDDS was emplaced mainly in four strike orientations i.e. NNE-SSW, WNW-ESE, N−S and NNW-SSE. The NNE-SSW and WNW-ESE trending members of NDDS are most abundant. The mafic dykes are represented dominantly by dolerites and related gabbroic variants although minor peridotites, norite and granophyric dykes have also been reported (Krishnan, 1936; Dunn and Dey, 1942; Saha et al., 1973). Mallik and Sarkar (1994) have classified the NDDS in three phases viz. at ~2100 ± 100, at ~1500 ± 100 Ma, and at 1100 ± 200 Ma (based on K-Ar ages of Mallik and Sarkar, 1994; Sarkar et al., 1969; Sarkar and Saha, 1977). In addition, Rb–Sr whole-rock isochron age of 2613 ± 177 Ma was reported for NNE–SSW trending ultramafic dyke (Roy et al., 2004). First precise Pb-Pb baddeleyite age (using analytical method of Kumar et al., 2014) of 1765.3 ± 1.0 Ma was reported on WNW-ESE trending members of NDDS (Shankar et al., 2014). Very recently a 2762.4 ± 2 Ma dyke swarm, 2750 ± 0.9 Ma dyke and 2800.2 ± 0.7 Ma dyke have been discovered on NNE-SSW trending members of NDDS (Kumar et al., 2017).

3. Sampling and methodology

We have targeted the WNW-ESE trending 1765 Ma dykes for this study. Most of the sampled dykes are vertical or sub-vertical and extend up to 200 km, with a thickness ranging from 14 m to 210 m (Fig. 1 and Table 1). Most of the sampled dykes are doleritic in composition. Almost all the samples for this study were collected from fresh exposures in stream and road cuttings. 86 oriented block samples were collected from 9 sites from WNW-ESE trending dyke swarm (Fig. 1) using Brunton magnetic compass and Sun compass. Each site, in this study, is representing a distinct dyke. The block samples were collected fairly apart from each other at each site. At least eight such block samples were collected from each site. In addition of this, host granite samples were also collected for baked contact test. Ten oriented block samples were collected from baked host granite at 30 cm away from dyke's chilled margin at Site K12 (Fig. 6(a)). Nine oriented block samples of host granite were collected 80 m away from dyke's chilled margin (Fig. 6(a)). Samples were later drilled and cut at rock cutting section, CSIR-National Geophysical Research Institute. A minimum of three cores with 2.56 cm diameter were drilled from each of the block samples. These cores were then cut into cylinders of 2.2 cm length. At last, we have 3–5 cylindrical specimens from each block sample. About 400 cylindrical specimens were obtained in this way and about 180 were subjected to following studies.

Standard demagnetization procedures including incremental alternating field (AF) and/or heat treatment (TH) were performed using Molspin Alternating Field demagnetizer and MMTD 80 oven (Magnetic Measurements Limited) respectively. At least one specimen per sample were subjected to detailed stepwise AF and/or thermal demagnetization, to a maximum of 120 mT or 620 °C. After each step of incremental demagnetization, the natural remanent magnetization (NRM) was measured using JR-6 spinner magnetometer (Agico). The directional data was analysed, using Principal Component Analysis (PCA) (Kirschvink, 1980) in combination with Zijderveld diagrams (Zijderveld, 1967). Acceptance criteria for linear segments on vector diagrams were the number of consecutive demagnetization points (≥5) and maximum angle of deviation ≤ 10° (Kirschvink, 1980). In most cases the stable characteristic remanence magnetization (ChRM) direction is defined by clear linear segments to the origin, which has been isolated by demagnetization due to either after 40–100 mT AF treatment or after heating at 400–600 °C (Fig. 3). Isothermal remnant magnetization (IRM) studies were performed on an ASC Scientific Model IM-10–30 impulse magnetizer.

4. Results

4.1. Petrography

The dyke samples are mostly doleritic in composition containing calcic-plagioclase and clinopyroxene (Fig. 2a–d). Theses samples are characterised by ophitic to sub-ophitic textures. The essential mafic minerals are pyroxenes, plagioclase and amphibole. Plagioclase was in the form of euhedral to subhedral tabular grains (Fig. 2c; 2d). Plagioclase varies in composition from anorthite to albite. Sericitization was observed in fine grained plagioclases (Fig. 2c; 2d). The clinopyroxene was mainly augite in the form of euhedral to subhedral prismatic grains with pinkish to greenish pleochroism and inclined extinction (Fig. 2c; 2d). Pyroxene in some altered samples is replaced by chlorite and actinolite. Alteration of both plagioclase and pyroxene has been attributed to hydrothermal activity (Sengupta et al., 2014). Amphibole was in the form of prismatic grains, showing greenish pleochroism. Accessory minerals are opaques (mainly magnetite), apatite, chlorite and rutile. Petrographic studies show that there is no major difference in the primary mineral compositions among samples from different dykes.

4.2. Paleomagnetism

The results of paleomagnetic studies from 9 sites are presented in Table 1 and also plotted in Figs. 3, 4 and 5. NRM intensities for samples range from 11.6 mA/m to 1 A/m. The higher coercivity component defines a well-grouped ChRM, with north to north-westerly declination and shallow negative inclination (Fig. 4). Within errors, all the sites

Table 1
Results of paleomagnetic measurements on the Singhbhum 1765 Ma WNW-ESE dyke swarm.

Site	Lat. (°N)	Long. (°E)	T	S°	N	D(°)	I(°)	k	α_{95}	Plat. (°N)	Plong. (°E)	dp	dm
K10	21.593342	85.726433	102	294	10	318.1	−10.0	167.9	3.7	41	328	1.9	3.8
K11	21.593239	85.743685	32	302	10	332.5	−35.1	56.1	6.5	41	301	7.5	8.5
K1	21.551608	86.017733	14	291	10	308.3	−21.8	482.3	2.2	30	328	1.2	2.3
K12	21.597614	85.746711	40	307	8	315.3	−14.9	48.5	8.0	37	327	4.2	8.2
Baked	21.597942	85.746536			10	319.7	−10.5	72.3	5.7				
Unbaked	21.599314	85.745983			9	352.4	−40.9	295.8	3.0				
K14	21.372461	86.147928	192	305	10	345.9	−32.4	45.3	7.3	48	287	4.6	8.2
K15	21.430211	86.181267	133	298	8	326.5	−22.4	128.6	4.9	43	314	2.8	5.2
K18	21.517099	85.911348	62	297	10	336.1	−20.3	249.7	3.1	50	305	1.7	3.2
K30	21.596853	85.885169	140	303	10	343.4	−21.0	65.3	6.0	54	294	3.3	6.3
SKJ10	21.549188	85.822280	210	300	10	339.4	−22.7	216.4	3.5	51	299	2.0	3.7
Mean	**21.533513**	**85.909172**			**86**	**329.2**	**−22.8**	**31.6**	**9.3**	**45**	**311**	**5.2**	**9.9**

Lat. and long are the latitude and longitude in degrees; T is the thickness in meters; S° is strike of the dyke at the respective sites. N = Samples studied from each site; D is the mean declination; I is the mean inclination; k is the precision parameter; α_{95} is the radius (°) of the 95% circle of confidence about the mean magnetization direction; Plat. and Plong. are the paleocoordinates of the pole; and dp, dm defines an oval of confidence within which there is a 95% probability that the true pole lies; all statistical parameters are based on sample means.

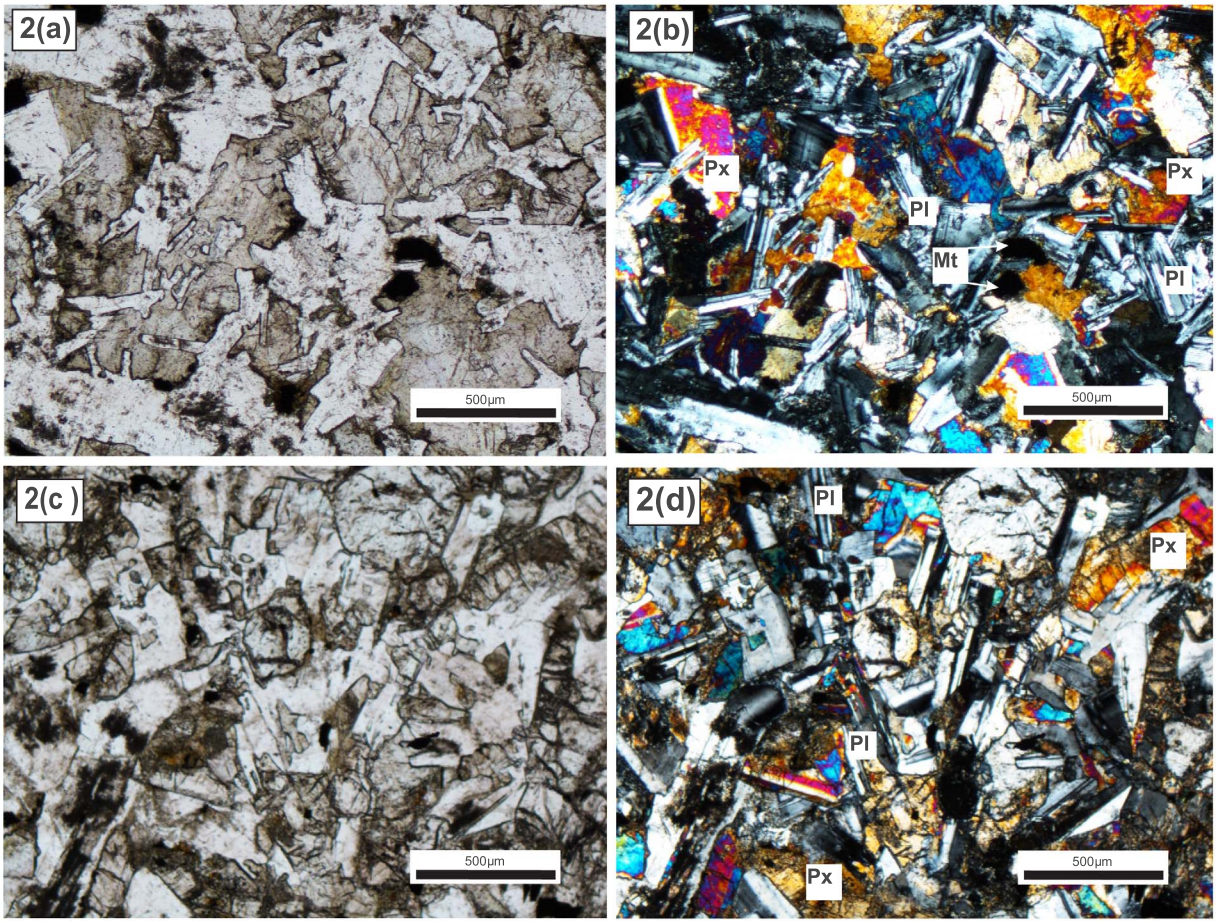


Fig. 2. Photomicrograph of WNW-ESE dyke samples. 2(a) and 2(b) are from site K11; 2(c) and 2(d) from site SKJ10.

regardless of their location have similar ChRM (Table 1 and Fig. 4). Mean ChRM was calculated with respect to the mean Site latitude: 21.533513°N and mean Site longitude: 85.909172°E. For all calculations and paleogeographic reconstructions only mean paleomagnetic data from this study have been used. Though there is published work on Paleomagnetism of NDDS (Verma and Prasad, 1974), we are not using those results here. The reason for not using the published data (Verma and Prasad, 1974) is the lack of reliability. It is not clear in the published work that whether the secondary component was completely removed or not. It is also not clear whether PCA was performed to achieve ChRM or not. No arguments have been presented to establish the primary nature of magnetic directions. Due to these drawbacks we are not using this data for any calculations. The WNW-ESE trending dykes has yielded a mean paleomagnetic direction with a declination = 329.2° and an inclination = −22.8° (k = 31.6; α95 = 9.3°) (Fig. 4). The pole position of SC at 1765 Ma is 45°N, 311°E (dp = 5.2 and dm = 9.9) (Fig. 5).

4.3. Baked contact test

We were able to obtain baked contact samples at site K12 during this study to prove the primary nature of the remanence directions (Fig. 6a–b). As mentioned in Section 3, ten oriented block samples were collected from baked host granite at 30 cm away from dyke (~40 m thick) chilled margin at Site K12 and nine oriented block samples were collected from baked host granite at 80 m away from dyke chilled margin Fig. 6c–e shows the demagnetization behavior of the dyke K12 and the host granite rocks in baked and unbaked zone. The mean ChRM direction for dyke K12 is declination = 315.3° and an inclination = −14.9° (k = 48.5; α95 = 8°). The mean ChRM direction for host granite in baked zone of dyke K12 is declination = 319.7° and an inclination = −10.5° (k = 72.29; α95 = 5.7°). The mean ChRM direction for host rock from unbaked zone of dyke K12 is declination = 352.4° and an inclination = −40.9° (k = 295.8; α95 = 3°). ChRM directions of the host granite rock in the baked contact zone of the dyke are similar to the ChRM direction of the dyke K12, and the ChRM directions of the unbaked host rock is significantly different from the dyke K12 directions (Fig. 6b). So we can see that the contact test is positive and hence the remanence direction is primary.

4.4. Rock magnetic study

Conventional IRM acquisition studies were also conducted on selected samples using ASC Scientific Model IM-10–30 impulse magnetizer. Typical saturation curves for these samples are shown in Fig. 7. The magnetic minerals in them saturate between 200 and 350 mT with coercivity between 30 and 70 mT. We have done REM (Remanent Efficiency of Magnetization) and $S_{-300\ mt}$ calculations, which are tabulated in Table 2. We can observe that most of the REM values (NRM/SIRM) are between 0.01 and 0.08, which suggest that the dykes have been cooled through curie temperature of magnetite which may be a carrier of magnetization (Wasilewski and Warner, 1988; Fuller et al., 1988). The $S_{-300\ mt}$ ($TRM_{-300\ mt}/IRM_{(saturation)}$) values are close to unity, which indicates the presence of magnetite (Basavaiah and Khadkikar, 2004). These rocks magnetic studies conducted on whole rock specimens, along with the petrography indicate that the dominant magnetic mineral is magnetite. The saturation field of magnetization (200–350 mT) along with coercivity values (30–70 mT) indicates that

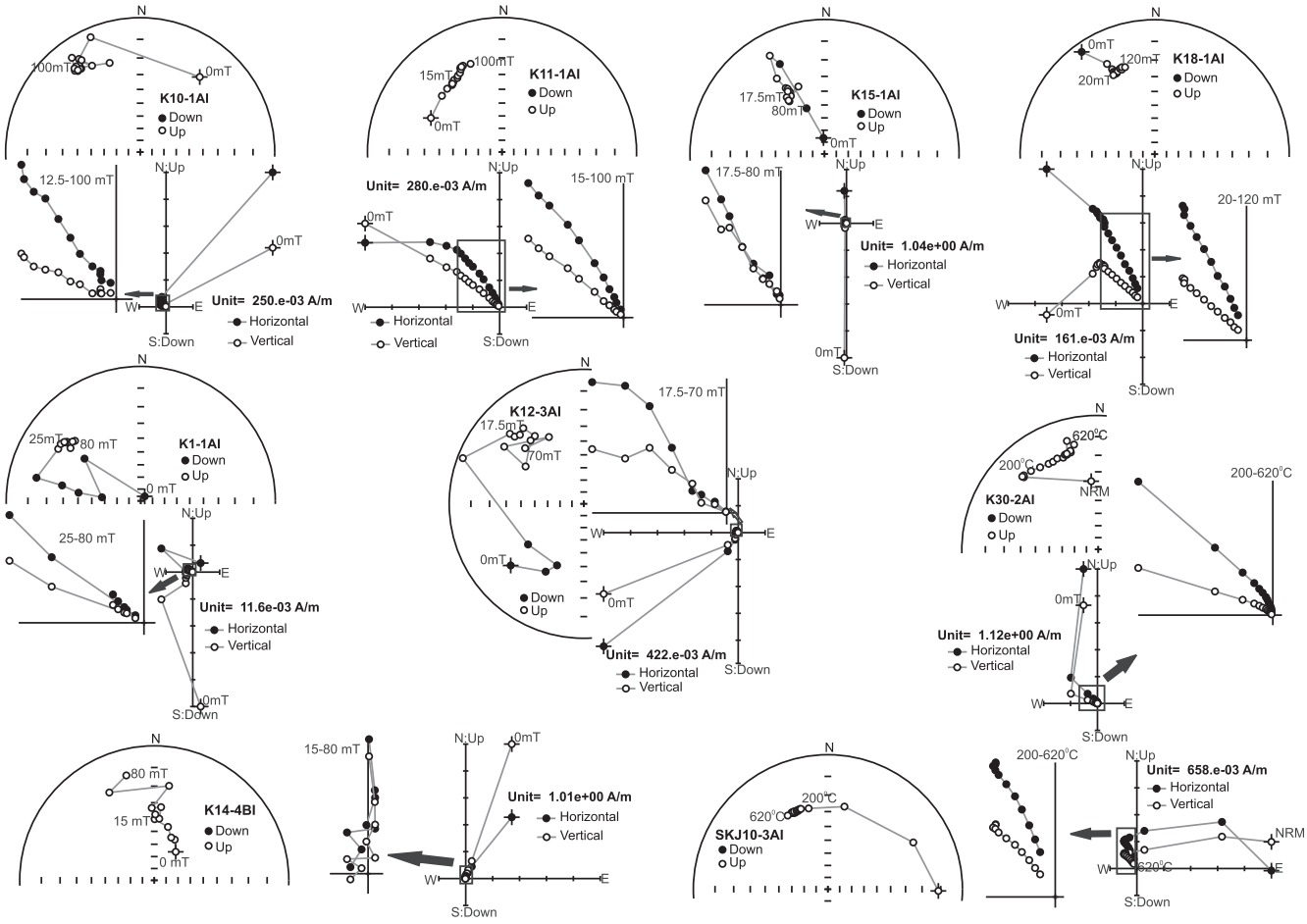


Fig. 3. Zijderveld diagram and equal area stereonet pairs showing the behavior of natural remanence to AF/Temperature for samples representing several sites. AF measurements are in millitesla (mT) and Temperature is in °C. Open/closed circles in the stereoplots represent upward/downward directed vectors and open/closed circles in the Zijderveld plots represent vertical/horizontal projections. Plotted using Remasoft 3.0 plotting and analysis program (Chadima and Hrouda, 2006).

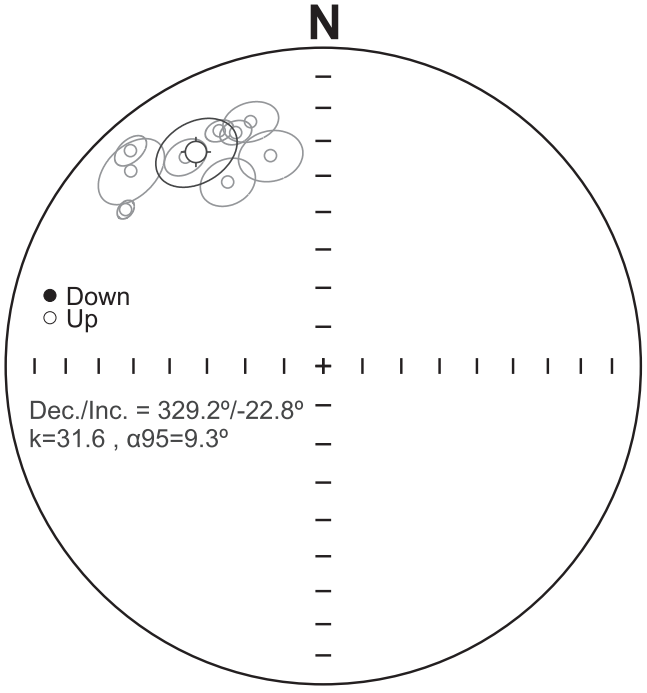


Fig. 4. Stereoplots showing paleomagnetic data. Primary site mean directions with ovals of 95% confidence of all the 9 WNW-ESE dyke samples and published sites. WNW-ESE dyke are shown 60% black and Mean direction is shown by black.

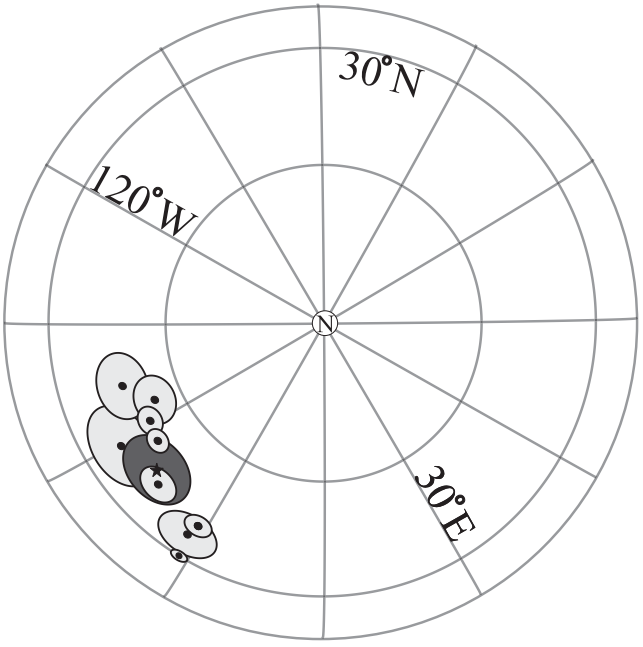


Fig. 5. Schmidt plot showing Mean Plaeopole position (black star) of 1765 Ma WNW-ESE trending Dyke swarm from Singhbhum Craton.

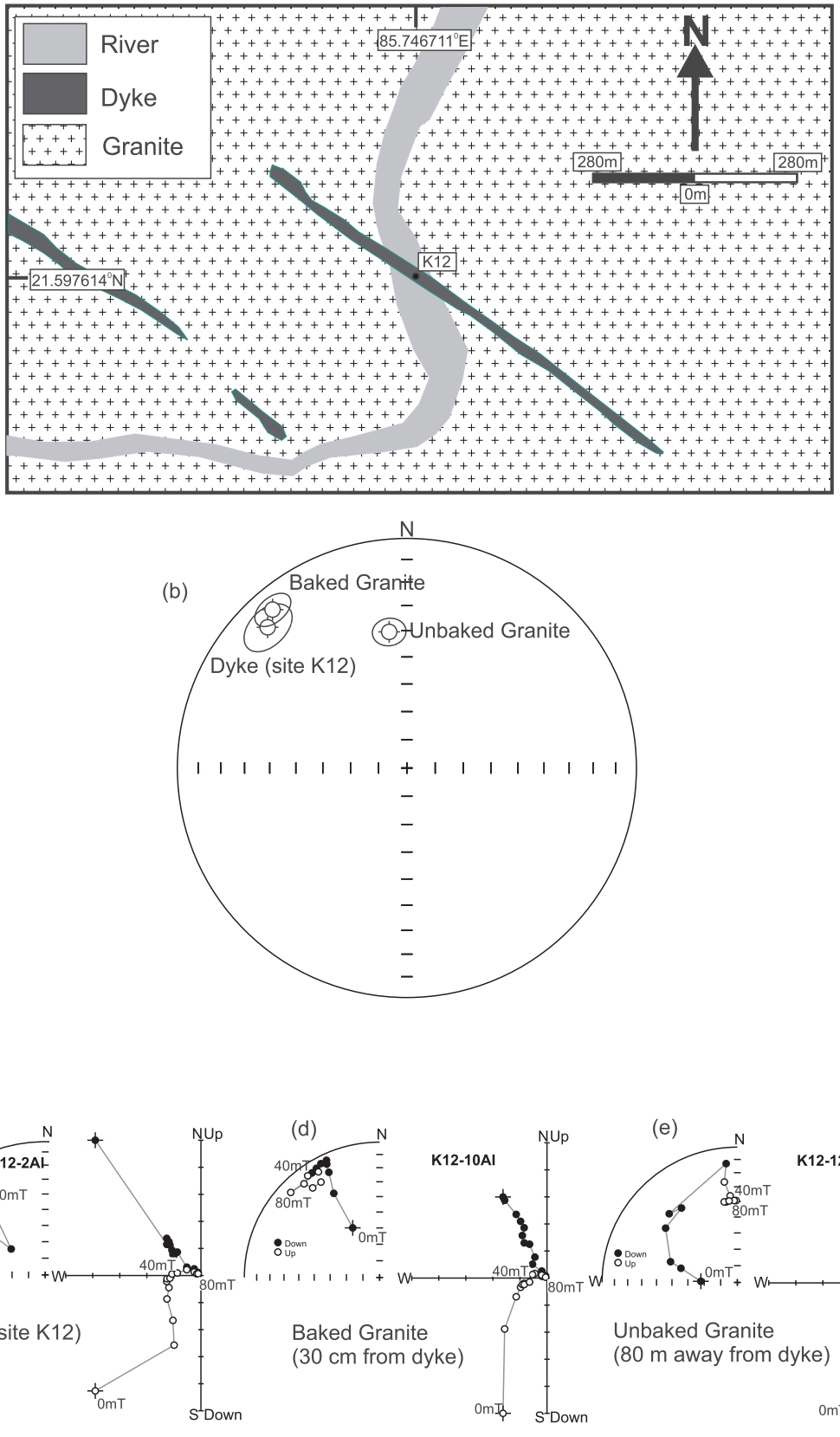


Fig. 6. Baked Contact Test at site K12. (a) Location map of Site K12, (b) Stereoplot for baked contact test at Site K12, and (c) to (e) Zijderveld diagram and equal area stereonet pairs showing the demagnetization behavior.

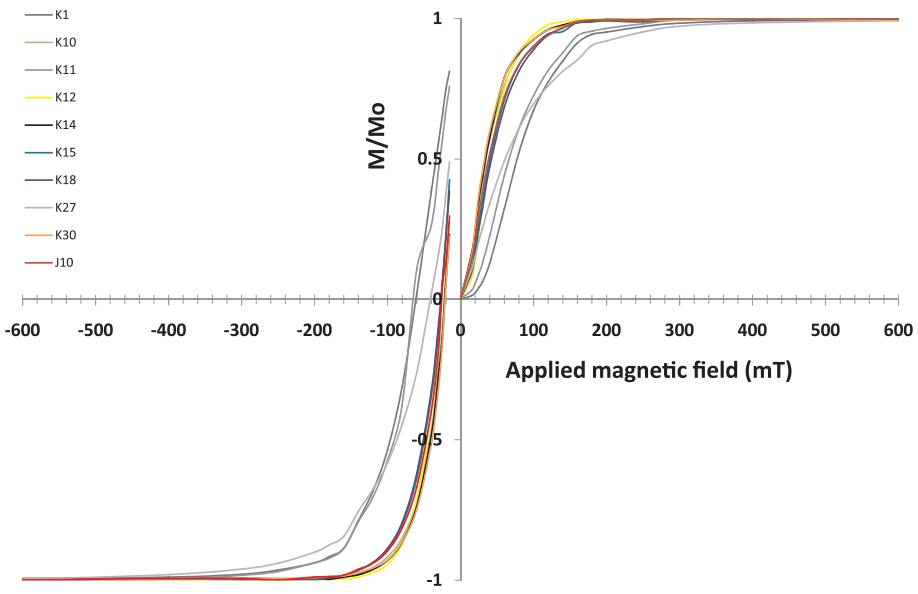


Fig. 7. Isothermal remanence acquisition curves (acquired magnezone normalized to saturated values vs. applied field in mT).

Table 2
Rock magnetic results of the selected dyke-samples.

Sample	NRM (A/m)	SIRM (A/m)	$IRM_{-300\ mT}$ (A/m)	REM	$S_{-300\ mT}$
K1	0.06	3.96	3.88	0.015	0.98
K10	1.51	133.06	133.13	0.011	1.01
K11	1.78	22.94	22.54	0.077	0.98
K12	1.51	447.10	445.11	0.003	0.99
K14	11.70	257.00	256.30	0.045	0.99
K15	5.34	82.64	81.81	0.064	0.98
K18	0.99	273.40	272.26	0.003	0.99
K27	2.97	25.12	23.93	0.118	0.95
K30	19.60	175.16	173.30	0.110	0.99

NRM: Natural Remanence Magnetization, SIRM: Saturation Isothermal Remanent Magnetization, $IRM_{-300\ mT}$: Isothermal Remanent Magnetization at −300 mT, REM: Remanent Efficiency of Magnetization, i.e., Ratio of NRM and SIRM, IRM: Isothermal Remanent Magnetization, $S_{-300\ mT} = IRM_{-300\ mT}/SIRM$.

the domain structure of magnetization is probably highly paleomagnetically stable Single Domain or Pseuso-single Domain magnetite rather than lower stable Multi Domain magnetite.

5. Discussion

5.1. Paleogeografic reconstruction at ~1770 Ma

Paleogeographic reconstructions of India with Baltica Craton and with North China Craton have been attempted by many workers in recent years, during Paleo-Mesoproterozoic. Some of the relevant reconstructions have already been mentioned in Section 1. The new key paleopole at 1765 Ma, of this study, can be used to perform a geologically satisfactory paleogeographic reconstruction using coeval poles from Baltica Craton and North China Crtaon (Table 3 and Fig. 8).

It is a common practice, while performing a paleogeographic reconstruction, to show continental/cratonic blocks in a single configuration, so that the blocks can be brought closest together within the permissible constraints of the geology and/or paleomagnetic data. It is to be noted that the paleolatitude and azimuthal orientation of each block are fixed by respective paleomagnetic data, but the paleolongitude and polarity are not defined (Buchan, 2014). Eight coeval paleopoles (Table 3) from Baltica Craton (5 paleopoles), India (1 paleopole) and North China Craton (2 paleopoles) have been used for the paleogeographic reconstruction in Fig. 8. The euler rotation parameters used in the abovementioned reconstruction is given in Table 4. This paleogeographic reconstruction was made using the free GPLATES software (http://www.gplates.org/).

Baltica Craton was assembled in its present day configuration at 1900–1700 Ma (Bogdanova et al., 2008; Elming et al., 2010). The Eastern and Western Blocks of North China Craton have collided along the Trans-North China Orogen at 1900–1800 Ma (Zhang et al., 2006) in the present-day layout. The Southern (Dharwar-Bastar-Singhbhum) and Northern (Aravalli-Delhi-Bundelkhand) cratons were assembled together at 2100–1900 Ma (Mohanty, 2010). So we are using the present day perimeter and shape for the Baltica Craton, India and North China Craton, for the paleogeographic reconstruction (Fig. 8).

This Baltica Craton – India – North China Craton reconstruction (Fig. 8) at ~1770 Ma, juxtaposes the western margin of India with southern part of Baltica in a way that Archaean Dharwar and Sarmatia cratons located next to each other (as suggested by Pisarevsky et al.,

Table 3
1800–1750 Ma paleomagnetic poles.

#	Rock Type	Age (Ma)	Plat (°N)	Plong (°E)	A_{95}(°)	Reference
1	Late. Svecofennian Rocks, Mean Baltica	~1800	49.6	221	10.8	Elming and Pesonen (2010)
2	Småland Intrusions Baltica	1784–1769	45.7	182.7	8.0	Pisarevsky and Bylund (2010)
3	Shoksha Formation Baltica	1780–1760	39.7	221.1	4.0	Pisarevsky and Sokolov (2001)
4	Ropruchey Sill, Karelia, Baltica	1754–1748	39.1	216.6	6.7	Fedotova et al. (1999) and Lubnina et al. (2012)
5	Hoting Gabbro Baltica	1760–1740	43	233.3	10.9	Elming et al. (2009)
6	Xiong'er Group North China	~1780	50.2	263	4.5	Zhang et al. (2012)
7	Taihang NNW dyke swarm, North China	1772–1766	36	247	2.8	Halls et al. (2000)
8	WNW-ESE Singhbhum Dyke Swarm, India	1766–1764	45	311	7.2	This study

Plat, and Plong = Latitude, and Longitude of Paleopole.
A_{95} = Ellipses of confidence at the 95% level about the Paleopole.

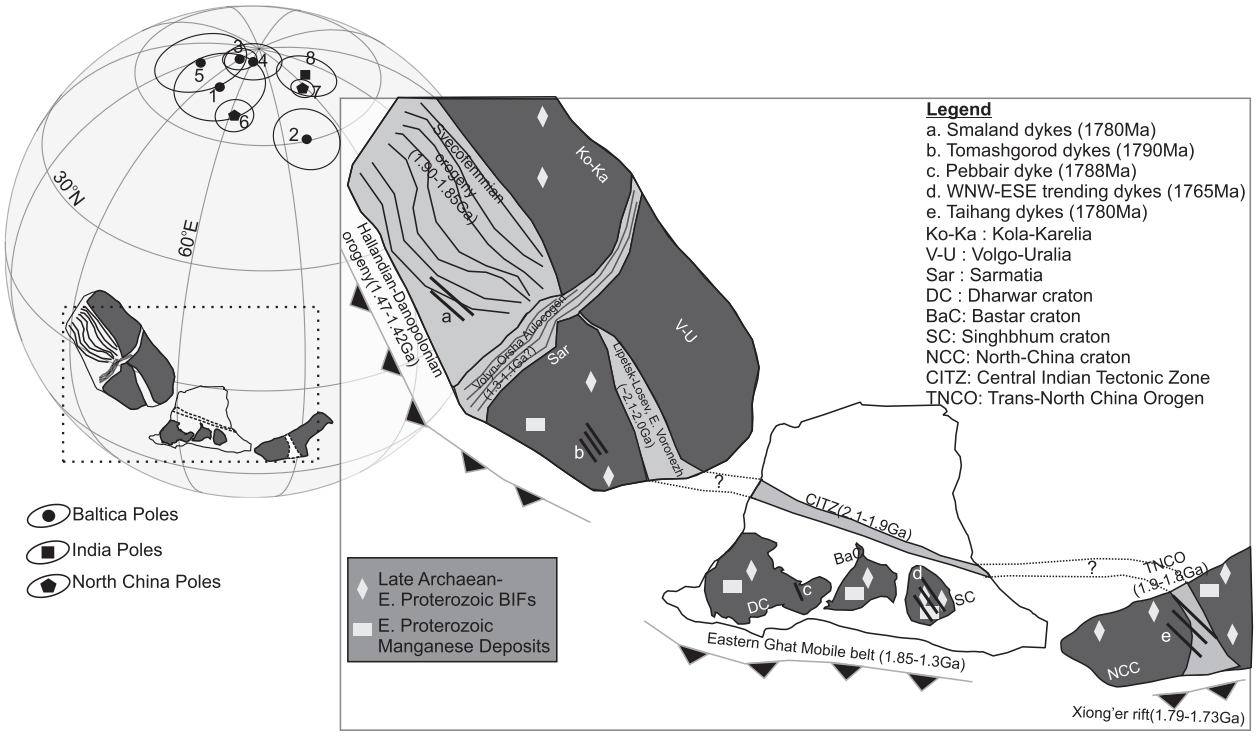


Fig. 8. Paleogeographic reconstruction at 1.77 Ga based on poles compiled in Table 3.

Table 4
Euler rotation parameters (to the absolute framework) for Fig. 8.

Craton/continent	λ (°N)	Φ (°E)	ω (°)
Baltica	−6.75	−47.76	−56.06
North China	17.44	−17.34	−65.72
India	14.46	37.84	−60.01

λ and Φ are the latitude and longitude of Euler poles respectively. ω is the Euler angle used for rotation. Euler angle (ω) is positive for counter-clockwise rotation and negative for clockwise rotation.

2013, 2014). In this position western block of North China Craton can be placed next to Eastern margin of India within permissible limits of Paleopoles.

5.2. Geological correlation for Baltica Craton – India – North China Craton reconstruction

The Baltica Craton, India and North China Craton are bounded by Paleoproterozoic orogenic belts (Fig. 8). The Lipetsk-Losev/East Voronezh Belt probably marks the 2100–2050 Ma accretionary orogen, which led to the collision with Volgo-Uralia with Sarmatia by 2020 Ma (Shchipansky et al., 2007; Bogdanova et al., 2008). The UHT metamorphism and deformation of almost similar age (2040 ± 17 Ma; Bhowmik et al., 2005) has been reported from the Central Indian Tectonic Zone (Mohanty, 2010). These tectonothermal events reflect some stage of amalgamation of the Dharwar/Bastar/Singhbhum and Bundelkhand/Aravalli cratons. The High pressure granulite facies event has been reported from Trans North China Orogen (TNCO) at 1915–1850 Ma (Zhang et al., 2006). This event suggest the amalgamation of Eastern and Western Blocks of North China Craton along the Trans-North China Orogen at 1.85 Ga (Zhao et al., 2004; Guo et al., 2002; Liu et al., 2002; Zhang et al., 2006 and references therein). Trends and positions and metamorphic ages of these three orogens in the present reconstruction (Fig. 8) suggest their possible genetic/spatial relationship.

The Smaland dykes (1780 Ma; Nilsson and Wikman, 1997; Pisarevsky and Bylund, 2010) from Fennoscandia peninsula; the Tomashgorod dykes (1790 Ma; Elming et al., 2010) from Sarmatia Craton; the Pebbair dyke (1788 Ma; Demirer, 2012) from Dharwar Craton; the WNW-ESE dykes (1765 Ma; Shankar et al., 2014) from SC; and the Taihang dykes (1780 Ma; Halls et al., 2000; Peng et al., 2005, 2006) from North China Craton, are aligned in an almost parallel trend (Fig. 8). The similarity in ages and the strike trends (in paleogeographic position) of these dykes (Fig. 8) indicate towards the possible genetic/spatial connection among them.

This Baltica Craton – India – North China Craton reconstruction (Fig. 8) aligns the southern accretionary margin of Baltica Craton with the eastern margin of India and the southern active margin of North China Craton. The presence of 1.85–1.33 Ga ophiolitic melange (Dharma Rao et al., 2011) in the Eastern Ghat mobile belt is suggestive of an active margin environment along the Eastern margin of India. The Baltica Craton's Proterozoic tectonic history is characterised by prolonged accretion from present west. This suggests a nearly linear long-lived Palaeo- to Mesoproterozoic 'Baltica-India' accretionary orogen (Pisarevsky et al., 2013; Fig. 8). The 1.79–1.73 Ga Xiong'er rift system (Peng, 2015 and references therein) is an indicator of a short lived active margin environment. The active margin environment is associated with subduction-related magmatism in India (Lakhna Dykes; Pisarevsky et al., 2013), in Baltica Craton (Post-Svecofennian crustal growth; Åhäll and Connelly, 2008) and in North China Crtaon (Xiong'er Volcanic Belt; Zhao et al., 2009). This suggest that Xiong'er rift system can be associated with the Baltica-India accretionary orogen (at least at ~1770 Ma; Fig. 8).

Late Archaean and Paleoproterozoic banded iron formations (BIFs) are widespread in Baltica Craton, Indian cratons and North China Craton (Fig. 8; Banerji, 1977; Khan and Naqvi, 1996; Shchipansky and Bogdanova, 1996; Srivastava et al., 2004; Li et al., 2015; Turchenko, 1992). Similarly the Early Proterozoic manganese deposits are reported from Sarmatia, Indian cratons, and North China Craton (Banerji, 1977; Li et al., 2015; Turchenko, 1992). The depositional model and environment of Paleoproterozoic metasedimentary iron-manganese deposits of India and North China Craton implied that the assembly may be located at low latitudes, where the conditions were favourable for

dissolving ice and precipitating manganese deposits (Li et al., 2015). The reconstruction proposed in Fig. 8 has constrained the paleoposition of Baltica Craton – India – North China Craton assembly at shallow latitudes. We understand that Late Archaean to Paleoproterozoic BIF and Paleoproterozoic Manganese deposits are found all over the world. But, the paleogeographic reconstruction (Fig. 8) presented here satisfy specific conditions (i.e., location at shallow latitudes for dissolving ice and precipitating manganese deposits) required for such deposits to form. And therefore the correlation of such deposits from Baltica Craton, Indian cratons and North China Craton gives an additional support for our reconstruction.

So we can argue that the paleogeographic reconstruction (Fig. 8) at ~1770 Ma is paleomagnetically permissible and supported by reasonable geological, tectonic and metallogenic evidences. We can propose here that the Baltica Craton and India linkage (proposed by Pisarevsky et al., 2013) can be extended back from ~1400 Ma to ~1770 Ma. At the same time we would also like to suggest that there is reasonable evidence in support of India-North China Craton spatial proximity at ~1770 Ma.

6. Conclusion

We are presenting here for the first time the paleomagnetic directions (declination = 329.2° and inclination = −22.8° with k = 31.6; α95 = 9.3°) from the 1765 Ma WNW-ESE trending dyke swarm from SC.

(1) We are reporting the first key paleopole for SC. The paleopole position of SC at 1765 Ma is 45°N, 311°E (dp = 5.2 and dm = 9.9).
(2) The paleogeographic reconstruction at ~1770 Ma, based on paleopoles (from Baltica Craton, India and North China Craton) and, coupled with a correlation of geological, tectonic and metallogenic features indicate towards an extended spatial link between Baltica Craton and India for ~370 Ma (1770–1400 Ma). There is reasonable evidence in support of India-North China Craton spatial proximity at ~1770 Ma.

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